The Stanley–Wilf Limit

A symbolic-method presentation and Lean 4 formalization

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Abstract

For every fixed permutation pattern τ, the sequence $|\mathrm{Av}_n(\tau)|^{1/n}$ has a finite limit. We present Arratia's supermultiplicativity argument [1] through the symbolic specification of an avoidance class as a sequence of indecomposable permutations; finiteness then follows from Marcus and Tardos [2]. We also describe a kernel-checked Lean 4 formalization of the full chain. Its extremal-matrix layer is an attributed Lean 4.29 adaptation of the Apache-2.0 `ForbiddenMatrix` development by Dillies and contributors [4]; this repository supplies Lean code for the permutation bridge, matrix-counting and dyadic arguments, symbolic route, and unconditional endpoint. This is a formalization and proof presentation, not a claim of mathematical priority.

Following Arratia [1], let $\sigma \oplus \pi$ denote the direct sum of two permutations and $\sigma \ominus \pi$ their skew sum. A nonempty permutation is *sum-indecomposable* (respectively *skew-indecomposable*) if it has no nontrivial decomposition using $\oplus$ (respectively $\ominus$).

Lemma 1. *Every nonempty permutation is sum-indecomposable or skew-indecomposable.*

Proof. If $\tau = \alpha \oplus \beta$ is a nontrivial direct sum, then the entry 1 occurs before the largest entry. If $\tau = \gamma \ominus \delta$ is a nontrivial skew sum, the largest entry occurs before 1. Both decompositions therefore cannot exist. □

Lemma 2. *If τ is sum-indecomposable, then $\mathrm{Av}(\tau)$ is closed under direct sum. If τ is skew-indecomposable, it is closed under skew sum.*

Proof. We prove the direct-sum case. Suppose that σ and π avoid τ. An occurrence of τ in $\sigma \oplus \pi$ either lies in one block, which contradicts avoidance by that block, or meets both blocks. In the latter case every selected value from the first block is below every selected value from the second block. The occurrence therefore expresses τ as a nontrivial direct sum, a contradiction. The skew case is identical with the inequalities reversed. □

Theorem 1 (Stanley–Wilf)**.** *For every permutation pattern τ, the limit*

$$\lim_{n\to\infty} |\mathrm{Av}_n(\tau)|^{1/n}$$

exists and is finite.

Proof. Write $c_n = |\mathrm{Av}_n(\tau)|$. Patterns of length at most one are immediate: all positive-size avoidance classes are empty, so the limit is zero. Assume $|\tau| \geq 2$.

By Lemma 1, choose direct sum or skew sum so that τ is indecomposable for that operation. Lemma 2 shows that $\mathrm{Av}(\tau)$ is closed under the chosen operation. Let $\mathcal{I}$ be its nonempty indecomposable members. Every component of an avoider is again an avoider, by hereditary pattern avoidance; conversely, closure reconstructs an avoider from such components. Unique indecomposable decomposition therefore gives the symbolic specification

$$\mathrm{Av}(\tau) = \mathrm{SEQ}(\mathcal{I}), \qquad C(z) = \frac{1}{1 - I(z)}.$$

Here $C(z) = \sum_{n\geq 0} c_n z^n$ and $I(z) = \sum_{n\geq 1} i_n z^n$ are formal ordinary generating functions. This is the unlabelled symbolic sequence construction [5]. In particular, concatenating the component sequences of objects of fixed sizes m and n gives an injection

$$\mathrm{Av}_m(\tau) \times \mathrm{Av}_n(\tau) \hookrightarrow \mathrm{Av}_{m+n}(\tau).$$

Indeed, all components have positive size, so the cumulative size m uniquely recovers the cut. Thus $c_{m+n} \geq c_m c_n$.

The singleton permutation avoids τ, so repeated sums give $c_n \geq 1$. Hence $\log c_n$ is superadditive. Fekete's lemma yields

$$\lim_{n\to\infty} \frac{\log c_n}{n} = \sup_{n\geq 1} \frac{\log c_n}{n},$$

initially allowing $+\infty$. Exponentiating gives

$$\lim_{n\to\infty} c_n^{1/n} = \sup_{n\geq 1} c_n^{1/n}.$$

Finally, the Marcus–Tardos theorem supplies a constant $K_\tau < \infty$ such that $c_n \leq K_\tau^n$ for all n. The displayed supremum is therefore at most K_τ. $\square$

Formalization

The Lean development formalizes both inputs rather than postulating the exponential bound. Its Marcus–Tardos extremal layer is adapted, with its Apache-2.0 license and modification notice, from Dillies and contributors' `ForbiddenMatrix` project [4]. For a pattern of length k, this layer gives the explicit Füredi–Hajnal linear bound

$$M_k = 2k^4 \binom{k^2}{k}.$$

The permutation-matrix embedding, a 2×2 block contraction, and the fifteen possible nonempty block masks give Klazar's recurrence [3], as reproduced in Marcus–Tardos [2, Section 3]. The bridge, recurrence, and dyadic argument are assembled here and prove the explicit bound

$$c_n \leq \left(2 \cdot 15^{2M_k}\right)^n.$$

The public Lean theorem has no mathematical hypotheses:

```
theorem stanleyWilf {k : Nat} (tau : Perm k) : GrowthTarget tau
```

where `GrowthTarget tau` states that the real sequence $c_n^{1/n}$ tends to a finite nonnegative real number, and `Perm k` is the type of permutations of $\mathrm{Fin}(k)$. The displayed supremum characterization belongs to the paper proof; it is not an extra field of `GrowthTarget`.

At the object level Lean constructs the canonical first-component bijection $\mathcal{C}_{n+1} \simeq \sum_j \mathcal{I}_{j+1} \times \mathcal{C}_{n-j}$ and derives the convolution and supermultiplicativity from it. The complete size-wise `SEQ` equivalence is then obtained from coefficient equality using `Fintype.equivOfCardEq`; it is not claimed to be an executable whole-factor-list decomposition.

Commit `cd7c4a7` builds with Lean 4.29.0 in a public CI run. A transitive `#print axioms` audit of 48 declarations reports only `propext`, `Classical.choice`, and `Quot.sound`. The source and verification scripts are available at github.com/xiangyazi24/stanley-wilf.

The mathematical argument is established work: Arratia supplies the supermultiplicative limit argument, Marcus and Tardos the extremal theorem, and their Section 3 records Klazar's counting reduction. The contribution of this artifact is its symbolic-method organization and verified Lean assembly.

References

[1] R. Arratia, *On the Stanley–Wilf conjecture for the number of permutations avoiding a given pattern*, Electron. J. Combin. 6 (1999), N1.

[2] A. Marcus and G. Tardos, *Excluded permutation matrices and the Stanley–Wilf conjecture*, J. Combin. Theory Ser. A 107 (2004), 153–160.

[3] M. Klazar, *The Füredi–Hajnal conjecture implies the Stanley–Wilf conjecture*, in *Formal Power Series and Algebraic Combinatorics*, Springer, 2000, 250–255, doi:10.1007/978-3-662-04166-6_22.

[4] Yaël Dillies and contributors, *ForbiddenMatrix*, Lean 4 formalization, Apache-2.0, GitHub commit `a938c04`.

[5] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.